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Course : CSA1104 - Object oriented Analysis and Code/Name Design

1, Architectural Patterns and MVC

Architectural patterns are reusable solutions for organizing the overall structure of an object oriented system.

Two common patterns:

- * Model - view - Controller (MVC)

- * Layered Architecture

MVC improves modularity by:

- * Model manages the data

- * View handles the user interface

- * Controller handles user requirement and coordinates both view and model.

2, Data Access object (DAO) Pattern

The DAO pattern provides the system access a separate object for performing database operations such as insert, update, delete.

Two Advantages:

- * Separates the database code for the object system.

- * Make database changes and testing easier.

3, Object - oriented Design Guidelines:

Object-oriented Design guidelines are principles used to create software that is reusable, maintainable, flexible and understandable.

Two Categories:

- * class design guidelines - focus on responsibilities, cohesion, encapsulation.
- * Relationship / design guidelines - focus on inheritance, coupling.

4. Database Persistence and ORM

Database persistence is the process of storing object data permanently so that it remains available after the application terminates.

Two Persistence Mechanisms:

- * Object - Relational Mapping (ORM)
- * JDBC / SQL - based persistence

5. Application Layer

The Application layer coordinates application use cases, and control the flow of operations.

Two major responsibilities:

- * Coordinates use cases
- * Manage communication between presentation and domain layers.

6. Three-Tier Architecture in E-Commerce

Three tier architecture consists of

Presentation Layer: User interface such as website / mobile app

Business layer: processes business rules and transactions

Data layer: Manage Database Operations.

7. Business Logic Layer:

The Business Logic Layer contains and processes the rules required by the application.

Two important functions:

- * Validate user requests.
- * Process business operations.

8. Cohesion and Coupling:

* Cohesion refers to how closely related the responsibilities within a class/module.

* Coupling refers to the level of dependency between different classes/modules.

9. Encapsulation and Data handling:

Encapsulation is the process of combining data and methods within a class and controlling access to the object data.

10. Layered Object-oriented system for Online Banking

Presentation layer: login, account details, fund transfer interface.

Application Layer: Coordinates operations such as transfer and balance enquiry.

Business/Domain Layer: contains banking rules, account validation, transaction processing.

11. Infrastructure Layer:

The infrastructure layer provides technical services required by the application and domain.

Two responsibilities:

- * Database Communication
- * Communication with external system

3. Singleton Design pattern

The Singleton pattern ensures that a class has only one instance and provides a global access point to the instance.

3. Dependency Injection:

Dependency Injection (DI) is a technique where an object's required dependencies are provided from outside instead of the object.

Two benefits:

- * Reduces tight coupling
- * Improves testability

4. Strategy Design Pattern

The Strategy Pattern defines a family of algorithms and allows them to be selected or changed at runtime.

Roles

- * Strategy
- * Concrete Strategy
- * Context

5. Layered Architecture for Library Management System

Presentation layer: login, book search, issue screens.

Application layer: Coordinates use cases such as use cases.

Business Layer: Contains rules for books, members, fines.